

David Wu

EE @ Cooper Union, interested in computers and mathematics.
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Education

Hunter College High School, New York, NY	Sep. 2020 - Jun. 2026
(Dual Enrollment) Hunter College, New York, NY	Sep. 2025 - Jun. 2026
- Multivariable Calculus, Linear Algebra, Intro to Data Science, Bayesian Stat	
Cooper Union for the Advancement of Science and Art, New York, New York	Sep. 2026 - Jun. 2030

Experience

Research Assistant | Distributed Artificial Intelligence Research (DAIR) Lab

Hunter College, Winter 2024 - Present

- Conducted research on multi-agent reasoning and collaboration between Large-language models for abstract reasoning by experimenting on harness architecture and prompts for ARC-AGI. Achieving significant improvements across the benchmark using collaboration.
- Gained hands-on experience in collaborative machine learning research environments, operating GPU clusters to train and finetune novel model architectures in classification and language models.
- Worked with a Masters student to write and submit a research paper to multiple AI conferences, contributing to experiment design, data analysis, and technical writing.

Software Engineering Intern | Atom Map Cooperation

New York, New York, Nov 2024 - Apr 2025

- Gained industry-level software development experience, including maintaining code quality, version control, and deployment best practices.
- Developed and deployed two custom Slack bot applications, actively used across the company's workspace to improve productivity by integrating with Zotero and Jiro workflows.
- Created and implemented custom command-line tools in Python to automate routine tasks using the OpenAI API, resulting in a 70% reduction in manual effort while creating pull requests

Software Engineering Intern | Teachshare

Harvard Undergraduate Ventures Tech Summer Program, Jun 2025 - Aug 2025

- Gained hands-on experience in the startup ecosystem through workshops and mentorship on product design, fundraising, and market validation.
- Interned at Teachshare, improving the backend infrastructure and enhancing AI implementation for automated worksheet generation.
- Designed and integrated new features into the AI worksheet creation system, increasing customization options and content accuracy for educators.

Private Tutor for Chess and Computer Science

In-person and online, 2021 - Sep 2025

- Provided 1:1 chess tutoring for 7 different students, focusing on strategy development, gameplay analysis, and tournament preparation.
- Tutored 4 students in computer science, covering topics from foundational programming concepts to advanced competitive programming/algorithms techniques.
- Designed and taught a group computer science course for 30+ students, incorporating algorithms, coding challenges, and project-based learning to build practical skills.

President | Vision Gavel Club (Toastmasters certified)

New York, New York, Feb 2025 - present

- Served as President of a Toastmasters-certified public speaking club, leading workshops and mentoring members.
 - ~12 members.
 - Organized and facilitated sessions in Chinatown, building communication and leadership skills for diverse participants.
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Projects

Speechly | Communication SaaS

<https://speechly.david3u.com>

Speechly is an application I originally made for my speech club to use AI to count the number of filler words from a recorded audio. From there, I kept adding more and more features to fully utilize the capabilities of AI and its multi-modal capabilities.

NYConductor | NYC public transport navigation assistant

<https://nyconductor.david3u.com>

NYConductor was created originally for my grandmother to use to navigate New York City's complex and English-centered transit system. It uses an AI agent with custom tooling connected to multiple different transit API endpoints to give the user answers to any question. The AI centric application allows for native questions and responses in all languages. The agent is also engineered to answer all non-traditional questions about the subway system, like finding which stops two lines intersect, or planning routes that accommodate stopping at a location.

Awards

- USACO Gold (non-certified platinum)
 - NYC DMI (Downtown Math invitational) team first place
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Skills

Programming Languages

- Python, C++, HTML, CSS, TypeScript, Java, SQL, R

Other Programming Skills

- AI workflows, MCP, OpenAI, PyTorch, Git, AWS, Pandas, Docker